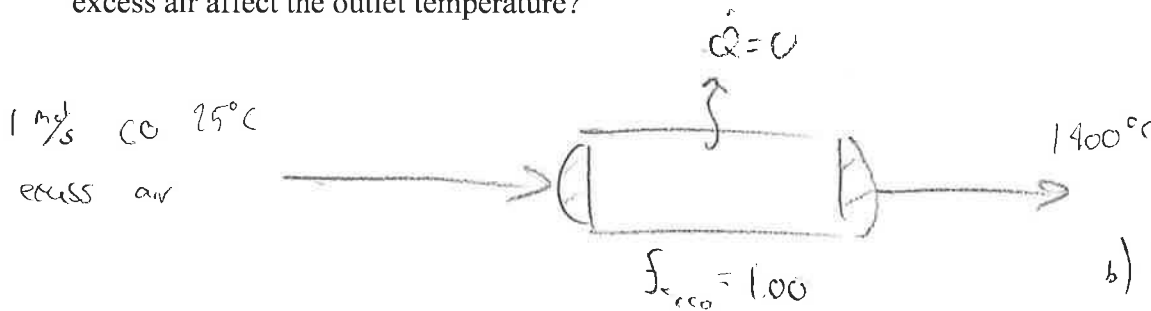
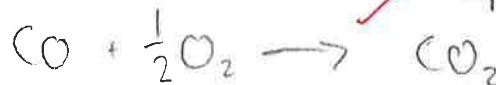


Carbon monoxide (CO) is to be burned with excess air in an adiabatic reactor. Assume 100% conversion of the CO and a basis of 1 mol/s of CO fed to the reactor. The reactants are fed at 25°C and the outlet gases exit at 1400°C. (a) Calculate the percentage excess air fed to the reactor. (b) How would increasing the percentage excess air affect the outlet temperature?



$$\Delta H_{CO}^0 = -283 \frac{\text{kJ}}{\text{mol}}$$



b) Increasing the percent excess air will cause the outlet temperature to decrease, because the heat released from combustion will remain the same w/ same CO flow rate, but now there are more molecules of air to soak up the heat, increasing the heat capacity of the mixture. Yes!

$$\text{Theoretical O}_2 \text{ moles} = \left(1 \frac{\text{mol CO}}{\text{s}} \right) \left(\frac{0.5 \text{ mol O}_2}{1 \text{ mol CO}} \right) = 0.5 \text{ mol O}_2$$

$$\text{Theoretical air (assume 21\% O}_2, 79\% \text{N}_2) = \frac{0.5 \text{ mol O}_2}{0.21 \text{ mol O}_2} = 2.381 \text{ mol air}$$

	n	$\hat{H}_f$	n	$\hat{H}_f$	
CO	1	0	0	0	
Air	n	0	$n - 0.5$	44.89	unknown air is n , and if 1 mol/s of CO is burned, then 0.5 mol O ₂ is depleted
CO ₂	—	—	1	71.89	

$$\% \text{ excess air} = \frac{5.2028 - 2.381 \text{ mol}}{2.381 \text{ mol}} \times 100\% = 118.5\%$$

$$\Delta H = 0 = [(n - 0.5) 44.89 + 71.89 + (-283)] \text{ kJ}$$

you forgot the N₂ in the air,

$$= 44.89n - 22.45 + 71.89 - 283$$

$$44.89n = 233.555$$

$$n = 5.2028 \text{ molar}$$

Table B.8 Specific Enthalpies of Selected Gases: SI Units

$\hat{H}(\text{kJ/mol})$							
Reference state: Gas, $P_{\text{ref}} = 1 \text{ atm}$, $T_{\text{ref}} = 25^\circ\text{C}$							
T	Air	O ₂	N ₂	H ₂	CO	CO ₂	H ₂ O
0	-0.72	-0.73	-0.73	-0.72	-0.73	-0.92	-0.84
25	0.00	0.00	0.00	0.00	0.00	0.00	0.00
100	2.19	2.24	2.19	2.16	2.19	2.90	2.54
200	5.15	5.31	5.13	5.06	5.16	7.08	6.01
300	8.17	8.47	8.12	7.96	8.17	11.58	9.57
400	11.24	11.72	11.15	10.89	11.25	16.35	13.23
500	14.37	15.03	14.24	13.83	14.38	21.34	17.01
600	17.55	18.41	17.39	16.81	17.57	26.53	20.91
700	20.80	21.86	20.59	19.81	20.82	31.88	24.92
800	24.10	25.35	23.86	22.85	24.13	37.36	29.05
900	27.46	28.89	27.19	25.93	27.49	42.94	33.32
1000	30.86	32.47	30.56	29.04	30.91	48.60	37.69
1100	34.31	36.07	33.99	32.19	34.37	54.33	42.18
1200	37.81	39.70	37.46	35.39	37.87	60.14	46.78
1300	41.34		.97	38.62	41.40	65.98	51.47
1400				41.90	44.95		56.25
1500				45.22	48.51		61.09

$$Q = \dot{m} \hat{H} = 1 \text{ kg/s} \times 94.189 \text{ kJ/kg} = 94.189 \text{ kW}$$

$$= 47.07 \text{ kW}$$

$$N = 99$$

$$CO_2 = 71.89 \text{ kg}$$